

Rapid Atmospheric Vapor Deposition of H:In₂O₃ Transparent Conducting Oxide Thin Films

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Abstract

Transparent conducting oxides (TCOs) are essential for the optoelectronics industry, but there is a critical gap in cost-effective methods to rapidly deposit low sheet resistance, high transmittance films without damaging delicate materials, including emerging soft semiconductors like metal-halide perovskites. In this work, atmospheric pressure chemical vapor deposition (AP-CVD) is used to synthesize $\text{H}:\text{In}_2\text{O}_3$ films with $7.20 \pm 0.01 \, \Omega \, \square^{-1}$ sheet resistance ($0.50 \pm 0.06 \, \text{m}\Omega \, \text{cm}$ resistivity) and transmittance up to 89% in the near-infrared (NIR), surpassing commercial sputter-deposited indium tin oxide. The growth rate is $40\times$ higher than atomic layer deposition (ALD), and the AP-CVD films are fully processed under atmospheric conditions at only $140 \, ^\circ\text{C}$. Comparison of secondary ion mass spectrometry and time-of-flight elastic recoil detection analysis with changes in carrier concentration indicate that H dopants are introduced from the water oxidant. There is an increase in mobility from $40 \pm 10 \, \text{cm}^2 \, \text{V}^{-1} \, \text{s}^{-1}$ to $160 \pm 30 \, \text{cm}^2 \, \text{V}^{-1} \, \text{s}^{-1}$ when changing from O_2 to H_2O as the oxidant, which is attributed to H dopants passivating oxygen vacancies that act as carrier scattering centers. This work establishes AP-CVD as a promising method for manufacturing high figure-of-merit TCOs in a rapid, scalable and cost-effective manner, using mild growth conditions compatible with thermally-sensitive materials.

Keywords: $\text{H}:\text{In}_2\text{O}_3$, transparent conducting oxide, atmospheric pressure chemical vapor deposition, thin film manufacturing, high-throughput deposition, near-infrared transmittance

Introduction

Transparent conducting oxides (TCOs) are vital components in a wide range of technologies, including photovoltaics (PVs), microelectronics, touch screen displays, smart windows and sensors [1-5]. In several next-generation applications, including flexible photovoltaics, organic electronics, and temperature-sensitive optoelectronic devices, deposition under mild, atmospheric conditions is highly desirable [6-9]. Polymer substrates and functional layers made from organic or hybrid materials often impose strict restrictions on deposition temperatures ($<150 \, ^\circ\text{C}$), while vacuum-based processing increases system complexity, can increase capital costs, and limits process flexibility. Indium tin oxide (ITO) is the most widely used TCO commercially, due to its high electrical conductivity and optical transmittance in the visible wavelength range [6-9]. However, although atmospheric-pressure and solution-based routes to depositing ITO have been demonstrated [10-13], these methods generally require processing temperatures of $\sim 300\text{--}600 \, ^\circ\text{C}$, highlighting the difficulty of achieving high electrical performance with low-temperature processing under atmospheric conditions. Developing high-mobility TCOs that can be deposited rapidly at low temperature without vacuum infrastructure, and without requiring harsh oxidizing conditions (*e.g.*, use of plasma), therefore addresses an important materials and processing gap.

All high-performance TCOs achieve low resistivity through a combination of high carrier concentration (n) and/or carrier mobility (μ), since $\rho = 1/(ne\mu)$. In ITO, Sn incorporation primarily increases carrier concentration. While effective in lowering resistivity, high carrier densities enhance free-carrier absorption (FCA), particularly in the near-infrared (NIR), and increase ionized impurity scattering, which can limit mobility [14, 15]. This trade-off between conductivity with NIR transmittance and mobility becomes increasingly relevant for applications requiring high transmittance beyond the visible spectrum, such as in telecommunications and night-vision devices [16, 17]. Hydrogen-doped indium oxide (H:In₂O₃) provides an appealing alternative, in which hydrogen acts as a shallow donor with minimal perturbation of the conduction band. This enables moderate-to-high carrier densities while preserving high mobility [18–20]. In addition, hydrogen can passivate compensating defects and oxygen-vacancy-related trap states, further reducing carrier scattering [21, 22]. Reported mobilities for H:In₂O₃ prepared by ALD or sputtering are typically in the range of $\sim 120\text{--}130\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$, highlighting its potential as a high-mobility TCO that does not require high carrier density to achieve low resistivity [18, 20, 23]. These intrinsic advantages make H:In₂O₃ particularly attractive for broadband transparent electrode applications.

Despite these material benefits, deposition strategies remain a bottleneck. Sputter deposition is currently a well-established route for H:In₂O₃ deposition, due to its ability to deliver low resistivity ($< 0.3\text{ m}\Omega\text{ cm}$) with high visible and NIR transmittance ($>80\%$) [24, 25]. Nevertheless, the energetic plasma environment associated with sputtering can be detrimental to soft or defect-sensitive materials, including hybrid perovskites and polymeric substrates. Atomic layer deposition (ALD) has also been widely explored for H:In₂O₃-based TCOs due to its precise control over film thickness and quality, even on complex substrates [20, 26, 27]. This is achieved through self-limiting reactions and the sequential injection of gases to a reaction chamber. The thickness increment per precursor/oxidant sequence is termed the growth per cycle (GPC, typically expressed in Å per cycle). Numerous studies have shown that In₂O₃ can be grown by ALD using a variety of precursors at relatively mild temperatures (70–250 °C [28]), and the precursors have low kinetic energy, such that they do not mechanically damage the underlying materials [29–31]. Various precursors, including InCl₃ [20, 27, 32–39] and cyclopentadienyl indium (InCp) [20, 27, 32–39], have been employed with different oxidants in ALD, such as H₂O/O₂, H₂O₂, O₂ plasma, and N₂O, yielding a GPC ranging from 0.06 to 1.7 Å per cycle. This inherently low growth per cycle is characteristic of ALD. When film thicknesses of several hundred nanometers are required to achieve low sheet resistance, deposition times can extend to several hours, limiting throughput [40].

Spatial atomic layer deposition (SALD) has been developed to overcome the throughput bottlenecks of ALD by separating precursor streams in space rather than in time [41–45]. In close-proximity SALD, the substrate is held parallel to the gas manifold, and moves continuously beneath isolated precursor zones separated by inert gas curtains. When the manifold–substrate spacing is sufficiently small

(typically $<10\text{ }\mu\text{m}$), gas intermixing is suppressed, and surface reactions remain self-limiting, analogous to conventional ALD, with the inert gas curtains taking the place of the purge steps. The primary speed advantage arises from eliminating vacuum purge steps rather than significantly increasing GPC [46–48]. However, if precursor intermixing occurs due to increased manifold–substrate spacing or insufficient gas separation, gas-phase reactions are enabled and the process transitions towards a CVD-like growth regime. For clarity, we distinguish between growth per cycle (GPC) and growth rate. GPC refers specifically to the thickness increment per self-limiting ALD sequence. In contrast, growth rate (nm min^{-1} or $\text{\AA min}^{-1}$) describes thickness accumulated per unit time and depends on both GPC and cycle duration. In CVD-like processes where growth is not self-limiting, thickness scales directly with exposure time, and the growth rate is therefore the most meaningful metric for comparing how rapidly depositions are carried out.

Conventional ALD of $\text{H:In}_2\text{O}_3$ using InCp and $\text{H}_2\text{O/O}_2$ has been reported to deliver high Hall mobilities of $138\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ after post-deposition crystallization at $150\text{--}200\text{ }^\circ\text{C}$ in flowing N_2 [20]. Nevertheless, the need for relatively high thermal budgets and prolonged post-deposition treatment remains a drawback for applications requiring rapid and low-temperature processing. Promising results for the development of In_2O_3 using SALD was recently demonstrated using a trimethylindium (TMI) precursor with a GPC of $(1.33\text{ }\text{\AA per cycle})$ [49]. In this case, ozone was used as an oxidant to enhance the surface kinetics of the oxidation half-cycle. The deposited thin films achieved a Hall mobility of $54.7\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ at $225\text{ }^\circ\text{C}$ deposition temperature. When integrated as the active layer in flexible thin-film transistors (TFTs), the In_2O_3 films demonstrated a good field-effect mobility of $69.8\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ [49]. Despite the promise of this approach for microelectronics, the use of relatively high growth temperatures, as well as the use of ozone may pose challenges for applications requiring milder, softer chemical processes.

In this work, we address these pressing gaps in the manufacture of TCOs by introducing the use of atmospheric pressure chemical vapor deposition (AP-CVD) for these materials. This is comprised of a close-proximity SALD reactor operating in CVD growth mode, and we focus on $\text{H:In}_2\text{O}_3$ as a test case. For this material, it is critical to control the oxidants used. This is because the choice of oxidant influences the growth rate, concentration of oxygen vacancies, and how the hydrogen dopant is introduced, which collectively dictate the electrical, optical and structural properties of the $\text{H:In}_2\text{O}_3$ films. Therefore, in this work, we systematically varied the oxidizing environments used in AP-CVD $\text{H:In}_2\text{O}_3$, comparing $\text{O}_2/\text{H}_2\text{O}$, $\text{N}_2/\text{H}_2\text{O}$, O_2/N_2 , and O_2 in the oxidant channels. Depositions took place under fully atmospheric conditions, and to ensure we develop a mild growth process, we focused on deposition temperatures of $100\text{--}140\text{ }^\circ\text{C}$, and avoided the use of ozone or plasma. By correlating oxidant chemistry with film morphology, defect structure, hydrogen content, and optoelectronic properties, we investigate the mechanism of hydrogen doping and how this dopant affects electronic and optical

properties. Furthermore, we benchmark these optoelectronic properties against commercial ITO and previously reported In_2O_3 grown with slower vacuum-based deposition methods.

Results and Discussion

A side view of the gas manifold used to deposit $\text{H}:\text{In}_2\text{O}_3$ thin films by AP-CVD is schematically illustrated in **Figure 1a**. This reactor design is described in detail in Ref. [50]. In this system, N_2 gas is bubbled through solid InCp powder heated at 40°C , and this carries metal precursor vapors to the central channel of the gas manifold (green channel in Figure 1a labelled ‘M’). This channel runs throughout the entire width of the manifold, and the precursor vapors are introduced from both sides. A narrow slit guides the precursors vapors from this header channel vertically down to the substrate oscillating beneath the manifold. Oxidants are introduced through adjacent channels, which are spatially separated from the metal precursor channel by inert gas curtains. Herein, we investigated H_2O vapor and O_2 gas as the oxidants, and compared the use of O_2 or N_2 as the carrier gas for these oxidants, *i.e.*, the four combinations tested were: $\text{O}_2/\text{H}_2\text{O}$, $\text{N}_2/\text{H}_2\text{O}$, O_2 and O_2/N_2 .

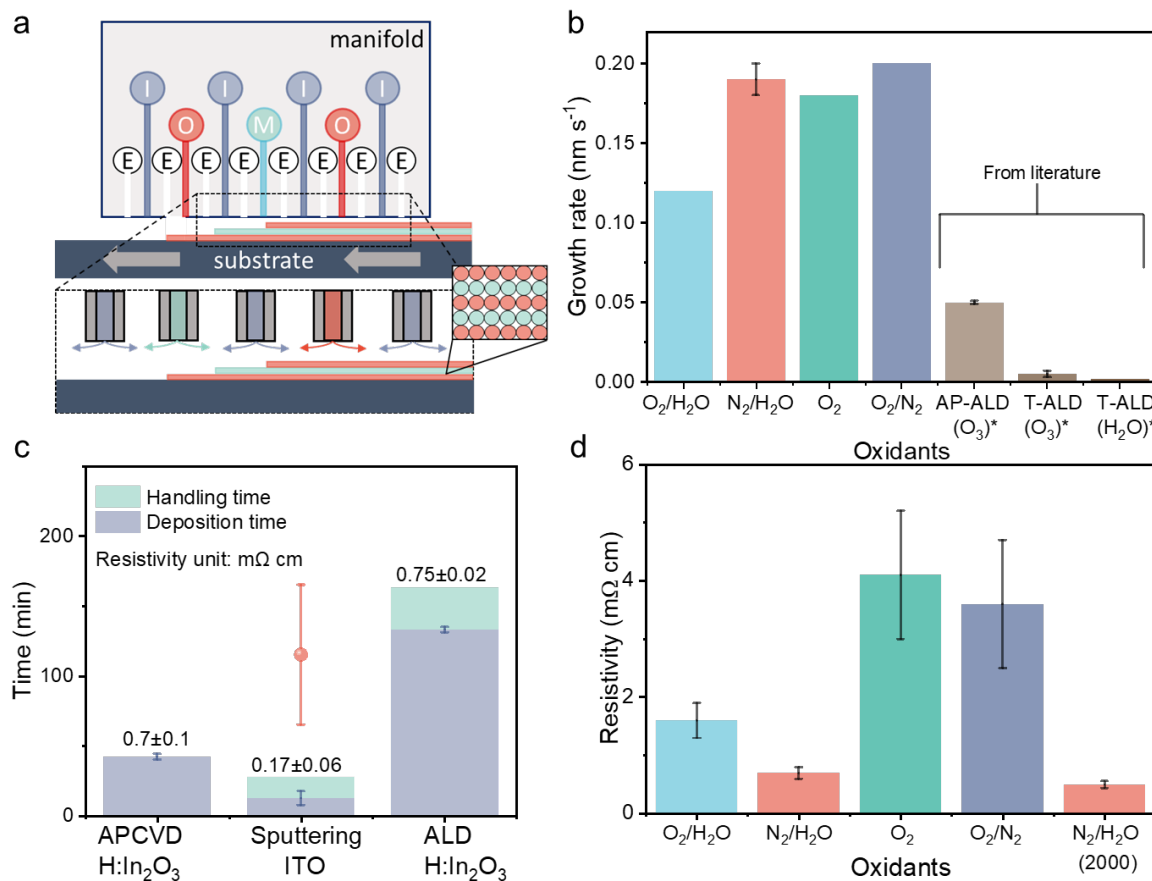


Figure 1. $\text{H}:\text{In}_2\text{O}_3$ grown by atmospheric pressure chemical vapor deposition (AP-CVD), compared with established alternatives. **a.** Illustration of the gas manifold used for AP-CVD, “M” denotes the metal precursor, “I” denotes the inert gas (N_2), “O” denotes the oxidant gas channels and “E” denotes the exhaust channels. The substrate is loaded onto a platen (metal plate), which heats the sample and oscillates it beneath the gas manifold. **b.** Growth rate of $\text{H}:\text{In}_2\text{O}_3$ thin films deposited by AP-CVD with

the four different oxidant combinations tested in this work, compared to the growth rate of H:In₂O₃ thin films by conventional ALD reported in the literature [49] (denoted as *). **c.** Total process cycle time to reach comparable bulk resistivity for sputtered ITO, ALD H:In₂O₃, and AP-CVD H:In₂O₃. Total cycle time includes deposition time and conservative estimates of vacuum handling (pump-down and venting), assumed to be 15 min for sputtering and 30 min for ALD. AP-CVD operates under fully atmospheric conditions and therefore requires no vacuum handling. The red outlier point represents sputtered ITO processes in which a post-deposition heat treatment step is included in the total processing time. Refer to **Table S2**, SI for details. **d.** Electrical properties of H:In₂O₃ thin films deposited at 140 °C for 1500 cycles, prepared using the four different oxidant combinations tested herein.

The deposition temperature is set by the heated platen that the substrate is mounted onto, which oscillates beneath the manifold, such that the number of oscillations determines the thickness of the film. In this configuration, one “cycle” corresponds to a single back-and-forth oscillation of the substrate beneath the gas manifold. During each oscillation, the substrate sequentially traverses precursor-rich and oxidant-rich regions. Although it is possible to set this manifold-substrate spacing to <10 μm and achieve ALD growth, in practice, a wider substrate-manifold gap is commonly used because of difficulties in maintaining perfect alignment between the gas manifold and substrate to the micron-level repeatedly across multiple samples. Increasing this spacing to >50 μm allows greater fluctuations in manifold-substrate spacing to be tolerated. However, this large spacing results in gas-phase mixing of the precursors, such that growth is by CVD rather than ALD. Nevertheless, prior work with AP-CVD has shown that similar film properties can be achieved in CVD or ALD growth mode, including high film uniformity, provided that the partial pressure of the precursor gases in their respective channels are the same across the width of the gas manifold, and that the manifold-substrate spacing is uniform [51, 52]. Furthermore, there is the added benefit that CVD results in a higher growth rate than ALD [40, 53]. For these reasons, we herein also used a relatively large manifold-substrate spacing of 100 μm for the growth of H:In₂O₃. We checked the growth mode by measuring the film thickness as a function of precursor exposure time (by varying the speed at which the substrate oscillates) and found a linear relationship (**Figure S1a**, SI). Given that we kept the number of oscillations constant at 1500, film thickness is proportional to the GPC. Therefore, the linear relationship between thickness and time, without saturation, is consistent with CVD rather than ALD growth. We therefore refer to this process as atmospheric pressure chemical vapor deposition (AP-CVD) rather than spatial ALD.

We prepared H:In₂O₃ films at 140 °C (refer to **Table S1**, SI) for 1500 cycles with the four different oxidant combinations, and estimated the growth rate per unit time from the thicknesses obtained and the total time taken for deposition (42-43 min). This assumes growth rates remain constant throughout the entire deposition process, and therefore provides an estimate of the overall growth rate. As shown in Figure 1b, these growth rates achieved by AP-CVD far exceeded those reported for H:In₂O₃ grown by atmospheric pressure conventional ALD (AP-ALD); note that trimethylindium (TMI) was used as

the metal precursor for ALD, rather than InCp [49]. Focusing on the H:In₂O₃ grown with N₂/H₂O in the oxidant channel, we confirmed the thickness to vary linearly with the number of cycles. The growth rate of $0.19 \pm 0.01 \text{ nm s}^{-1}$ was $40\times$ higher than that of H:In₂O₃ deposited using conventional ALD ($\sim 0.005 \text{ nm s}^{-1}$) [27].

The high growth rates achievable with AP-CVD directly translate into reduced processing times. Figure 1c presents a comparison of the total processing time required to prepare H:In₂O₃ by AP-CVD *vs.* ALD compared to conventional sputter deposition of indium tin oxide (ITO). To make a fair comparison, we determined the processing times required to produce TCO films with similar resistivities (0.2–0.8 mΩ cm, refer to Table S2, SI) rather than the same thickness, since the doping levels in ITO and H:In₂O₃ are different. For sputtered ITO, we considered two common industrial workflows: (i) heated-substrate sputtering, where the substrate is held at elevated temperature during deposition (commonly $>250\text{--}300^\circ\text{C}$) to promote *in-situ* film crystallization and optical/electrical quality [54, 55], and (ii) low-temperature deposition followed by post-deposition annealing [56, 57]. In both cases, the deposition time is similar, and on the order of $25 \pm 5 \text{ min}$ for the film thickness considered, as shown in Table S2, but the effective total cycle time differs because of the annealing/thermal budget. Whilst sputtering benefits from a high deposition rate, the particles deposited would typically have high kinetic energy that could mechanically damage soft organic and hybrid materials (*e.g.*, lead-halide perovskites or organic photovoltaics [51, 58]). By contrast, ALD is a softer deposition method, but, as shown in Figure 1c, requires $>5\times$ longer processing time ($>160 \text{ min}$) to produce H:In₂O₃ with similar resistivity. AP-CVD is capable of bridging the gap between these two methods, combining the ability to deposit films in a soft manner, while also having a short processing time ($42 \pm 2 \text{ min}$) that is similar to that for sputter deposition. Notably, AP-CVD has negligible handling time, given that processing entirely takes place in atmospheric conditions, without requiring a vacuum chamber, in contrast to sputter deposition or ALD. This comparison highlights the combined advantage of high deposition rate, low deposition temperature and simplified atmospheric operation for AP-CVD compared to the well-established sputter deposition and ALD techniques.

Table 1. Comparison of growth rate, resistivity, sheet resistance, carrier density, and mobility of commercial ITO versus H:In₂O₃ thin films deposited with various oxidant combinations at 140 °C. c-ITO was purchased from Kintec Company without any treatment, while all H:In₂O₃ were prepared by ourselves for this work by AP-CVD. H:In₂O₃ films were deposited with 1500 cycles, unless otherwise specified. The growth per cycle and growth rate data for the c-ITO were not provided by the manufacturer, and so these data are listed here as “not reported” (n.r.). Carrier density uncertainty here was estimated by propagating independent uncertainties in thickness, Hall voltage fitting (typically within 5% [59]), and magnetic field calibration (3% calibration error), using standard root-sum-square error propagation. Mobility uncertainty was obtained by combining resistivity and carrier density uncertainties in quadrature.

Deposition condition	Thickness (nm)	Growth per cycle (nm)	Growth rate (nm s ⁻¹)	Resistivity (mΩ cm)	Sheet R (Ω □ ⁻¹)	Carrier density (10 ¹⁹ cm ⁻³)	Mobility (cm ² V ⁻¹ s ⁻¹)
c-ITO	150±3	n.r.	n.r.	0.19±0.02	13±1	122±8	27±3
O ₂ /H ₂ O	320±30	0.20	0.12	1.6±0.3	64±3	5.2±0.6	60±10
N ₂ /H ₂ O	470±40	0.36±0.01	0.19±0.01	0.7±0.1	14.3±0.5	5.7±0.6	160±30
O ₂	500±40	0.33	0.18	4±1	80±13	2.0±0.2	80±20
O ₂ /N ₂	560±50	0.37	0.20	4±1	65±12	4.1±0.4	40±10
N ₂ /H ₂ O (2000 cycles)	710±50	0.36±0.01	0.19±0.01	0.50±0.06	7.20±0.01	6.4±0.6	190±30

The electrical properties of the AP-CVD films deposited at 140 °C for 1500 cycles are displayed in Figure 1d and Table 1. Films grown with N₂/H₂O in the oxidant channel exhibited the lowest resistivity amongst all combinations of oxidants investigated, reaching 0.7±0.1 mΩ cm (1500 cycles, 470±40 thickness) with a Hall mobility of 160±30 cm² V⁻¹ s⁻¹. With an increase in thickness (2000 cycles, 710±50 thickness), the resistivity decreased further to 0.50±0.06 mΩ cm, while the mobility increased to approach 190±30 cm² V⁻¹ s⁻¹. In contrast, using O₂ in the oxidant channel (O₂ only, or O₂/N₂) resulted in higher resistivity (~4 mΩ cm for 1500 cycles), while using O₂/H₂O led to an intermediate resistivity (1.6±0.3 for 1500 cycles). These results indicate that oxidant chemistry strongly influences the electrical properties of these AP-CVD-grown films.

The optical transmittance of H:In₂O₃ thin films deposited under different oxidizing environments in the visible region is shown in Figure 2a. Spectra were measured for film-on-glass samples using a bare glass substrate as reference (films thicknesses stated in caption). Given that film transmittance depends on its thickness, caution is required when comparing films of different thicknesses directly. Despite being significantly thicker (470±40 nm) than the O₂/H₂O-grown film (320±30 nm), the N₂/H₂O-grown film exhibits comparable or higher visible transmittance, indicating favorable intrinsic optical properties. The difference becomes more pronounced in the NIR region (Figure 2b), where the optimized N₂/H₂O film deposited for 2000 cycles (710±50 nm thickness) exhibited an average NIR transmittance of

~89.0%, whereas commercial sputtered ITO (150 ± 3 nm thickness) has only ~67.8% transmittance in the same wavelength range. Unlike ITO, which relies on high carrier concentrations that increase FCA, H:In₂O₃ achieves comparable conductivity through higher mobility at more moderate carrier concentrations, thereby suppressing infrared absorption [14, 15]. To further decouple thickness effects, ellipsometry-derived optical constants were examined (Figure S2, SI). In the visible wavelength range (400–700 nm), the extinction coefficient (k) remains near zero for both films. In contrast, ITO exhibits a marked increase in k in the NIR, consistent with strong FCA. The N₂/H₂O-grown H:In₂O₃ film maintains a low extinction coefficient across the NIR region, confirming intrinsically suppressed NIR absorption. The reduced FCA is consistent with achieving low resistivity through high mobility (190 ± 30 cm² V⁻¹ s⁻¹) with moderate carrier density ($6.4\pm0.6 \times 10^{19}$ cm⁻³), rather than through extremely high carrier concentrations as in ITO ($1.22\pm0.08 \times 10^{21}$ cm⁻³). To elucidate the electronic structure of the best-performing N₂/H₂O-grown H:In₂O₃ thin film, ultraviolet photoelectron spectroscopy (UPS) was performed to determine the valence band maximum (VBM) as well as work function (Φ) (Figure S3a-b, SI). The optical bandgap, obtained from UV-Vis spectrophotometry (**Figure 2a**), was combined with these values to construct the full band structure (Figure S3c, SI). The Fermi level is positioned close to the conduction band minimum (CBM), indicating degenerate *n*-type behavior consistent with hydrogen-induced shallow donor states.

Finally, we note that whilst the resistivity could be reduced through further increases in film thickness, when we increased the number of cycles of AP-CVD H:In₂O₃ beyond 2000 cycles, the transmittance decreased, with only marginal improvements in conductivity (Figure S3, SI). At a deposition temperature of 140 °C, and with N₂/H₂O in the oxidant channels, we found 2000 cycles to strike the optimal balance between conductivity and transmittance. For all subsequent characterization, we focused on the films with thicknesses shown in Table 1, unless otherwise specified.

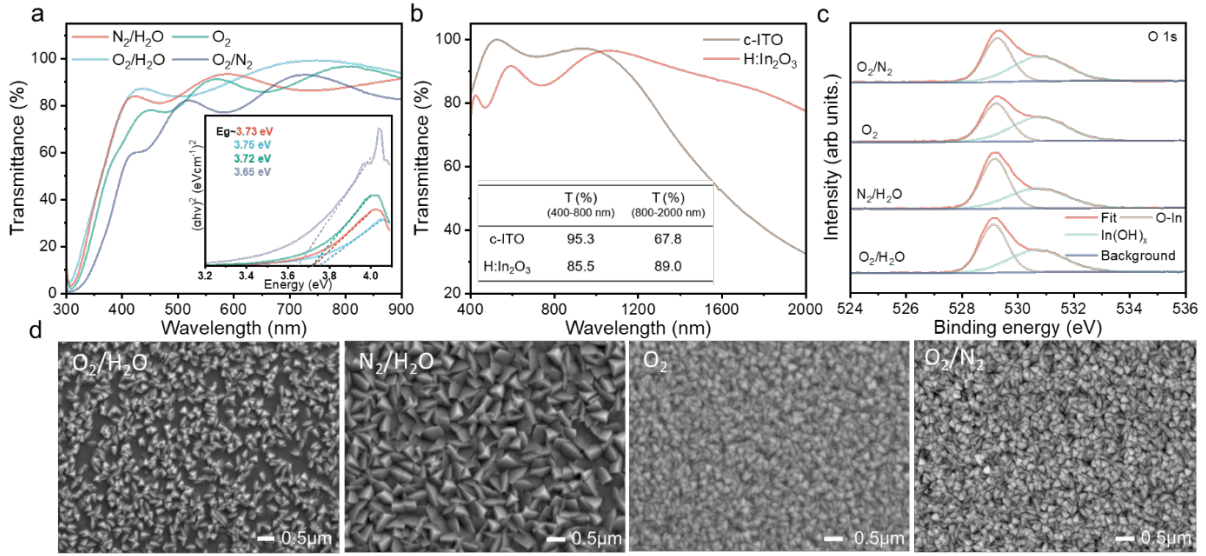


Figure 2. Effect of oxidant on optical properties, surface chemistry and morphology of AP-CVD H:In₂O₃ films. **a.** Transmittance and bandgap of H:In₂O₃ thin films deposited via AP-CVD at 140 °C for 1500 cycles under various oxidizing environments. The thicknesses of film grown with O₂/H₂O, N₂/H₂O, O₂ alone and O₂/N₂ in the oxidant channel are 320±30, 470±40, 500±40, 560±50 nm, respectively. **b.** Visible-IR transmittance of commercial ITO (150±3 nm thickness) and optimal N₂/H₂O-grown H:In₂O₃ thin film (710±50 nm thickness) deposited at 140 °C for 2000 cycles. **c.** XPS spectra, and **d.** high-resolution SEM images of H:In₂O₃ thin films deposited under various oxidizing environments for 1500 cycles.

The X-ray photoelectron spectroscopy (XPS) survey spectra of the In₂O₃ films (**Figure S5a**, SI) reveal indium and oxygen signals in all samples (details in Supplementary Note 1 and **Figure S5a**, SI), confirming the formation of In₂O₃ [60-62]. The O 1s spectrum reveals two distinct peaks (**Figure 2c**): O_I at ~529.2 eV, corresponding to In–O–In bonding, and O_{II} at ~530.8 eV, associated with oxygen vacancies [60, 61, 63]. The relative intensity of the O_{II} peak varies across different oxidant environments, indicating different levels of oxygen vacancy concentration [60, 61, 63]. The N₂/H₂O-deposited film exhibits the lowest O_{II} peak area, signifying the lowest oxygen vacancy formation, whereas the O₂-only film shows the highest O_{II} peak area, indicating a greater density of oxygen vacancy-related defects. The In 3d core-level spectra (**Figure S5b**, SI) further supports this observation, as illustrated in **Table S3**, SI. Although it may seem counterintuitive that films deposited under O₂-only conditions exhibited a higher vacancy-related XPS peak, high oxygen partial pressure during AP-CVD can promote rapid precursor combustion and non-equilibrium surface reactions, potentially leading to structural disorder and defect formation. In contrast, using N₂/H₂O in the oxidant channel may be conducive towards more moderate oxidation kinetics, and facilitate controlled hydrogen incorporation, thereby suppressing oxygen vacancy formation and enabling defect passivation. Hydrogen incorporation acts as a shallow donor and can passivate deep oxygen-vacancy-related states, as well as neutralize compensating acceptor-type defects, such as indium vacancies. The relevant defect reactions and energetic

considerations are summarized in Supplementary Note 1. These hydrogen-mediated processes could reduce vacancy-related disorder and contribute to improved carrier transport.

Structural characterization further supports this interpretation. XRD patterns (**Figure S4**, SI) confirm the cubic bixbyite phase (space group: $Ia\bar{3}$, No. 206) for all films. The N_2/H_2O -grown film exhibits the sharpest and most intense (222) reflection, indicating a higher degree of crystalline ordering. In contrast, O_2 -only films with similar or even higher thicknesses show broader reflections and mixed orientations, consistent with increased structural disorder. High-resolution SEM images (Figure 2d) reveal corresponding microstructural differences. The N_2/H_2O films exhibit the largest grain size and most compact morphology, whereas O_2 -containing conditions produce smaller grains and more porous microstructures. Larger grains reduce grain-boundary density and are expected to decrease carrier scattering. These structural trends correlate directly with the electrical data presented in Table 1. The N_2/H_2O -grown films, which exhibit the largest grains and highest crystalline ordering, also demonstrate the highest mobility ($160\pm30\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$) and lowest resistivity ($0.7\pm0.1\text{ m}\Omega\text{ cm}$). Reduced grain-boundary scattering and lower oxygen-vacancy-related disorder provide a structural and chemical basis for the enhanced transport properties and reduced infrared optical losses using N_2/H_2O [21, 64, 65].

Although figures of merit (FoMs) are often used to benchmark transparent conductors, their values depend strongly on film thickness and the spectral window chosen for averaging transmittance. In this work, FoM calculations (Figure S6-7 & Table S4, SI) indicate that the optimized $H:\text{In}_2\text{O}_3$ films are comparable to widely-used TCOs when evaluated over the visible wavelength range. However, because conventional FoM formulations typically exclude the NIR region, they do not fully capture the infrared transparency advantage offered by AP-CVD $H:\text{In}_2\text{O}_3$ (Supplementary Note 2). Accordingly, we emphasize thickness-independent parameters, such as resistivity and mobility when comparing deposition conditions. A broader comparison with literature-reported ITO, In_2O_3 , and $H:\text{In}_2\text{O}_3$ films deposited by various techniques is summarized in Table S5, SI. It is important to distinguish between the intrinsic advantages of $H:\text{In}_2\text{O}_3$ as a material and the advantages of AP-CVD as a deposition method. The high mobility and enhanced near-infrared transmittance originate from the $H:\text{In}_2\text{O}_3$ system itself, whereas AP-CVD offers a route to realize these properties using low-temperature, atmospheric-pressure processing. In this respect, AP-CVD compares favorably with ALD by delivering similar $H:\text{In}_2\text{O}_3$ film properties at much higher growth rates. AP-CVD is also favorable compared to sputtering by achieving similarly low resistivities and even higher mobilities without the use of plasma, thereby providing a milder process for sensitive substrates and device stacks.

To elucidate the mechanism of hydrogen incorporation in $H:\text{In}_2\text{O}_3$, it is essential to distinguish between hydrogen originating from the water oxidant versus atmospheric exposure or residual precursor ligands (e.g., Cp-derived C–H species). To address this, we performed isotope-labelling experiments by replacing H_2O with deuterium oxide (D_2O) during deposition, enabling discrimination between

hydrogen incorporated from the oxidant source and hydrogen introduced from extrinsic sources such as the atmosphere and precursors. It should be noted that although isotopic substitution from H_2O to D_2O may introduce minor kinetic isotope effects due to differences in O–H and O–D bond energies, the primary objective of these experiments is to identify the origin and depth distribution of hydrogen. The compositional trends observed here therefore support conclusions regarding hydrogen incorporation mechanisms, independent of potential small differences in reaction kinetics.

Nano-secondary ion mass spectrometry (Nano-SIMS) analysis was first conducted to map the depth-dependent deuterium-to-hydrogen (D/H) ratio (**Figure 3a**). As a control experiment, a film deposited under O_2 -only conditions at 100 °C for 1500 cycles exhibited a D/H ratio of ~ 0.0005 , consistent with the natural isotopic abundance of deuterium in atmospheric hydrogen [66, 67]. Note that we had to reduce the deposition temperature from 140 °C to 100 °C in order to obtain more amorphous films that presented a smoother sample surface for analysis, which is required for SIMS. Our measurements confirm that hydrogen detected using O_2 -only as the oxidant originates primarily from background contamination or residual Cp-related fragments (Figure S8, SI). In contrast, when the thin film was deposited with $\text{N}_2/\text{D}_2\text{O}$ in the oxidant channel at 100 °C for 1500 cycles, the D/H ratio exhibited a clear depth-dependent trend. Near the film surface, the D/H ratio was approximately 0.2, and increased to approximately 0.25 near the film/substrate interface. This D/H ratio is over 2.5 orders of magnitude greater than the film deposited without D_2O , providing strong evidence for the incorporation of hydrogen from the water-based oxidant used.

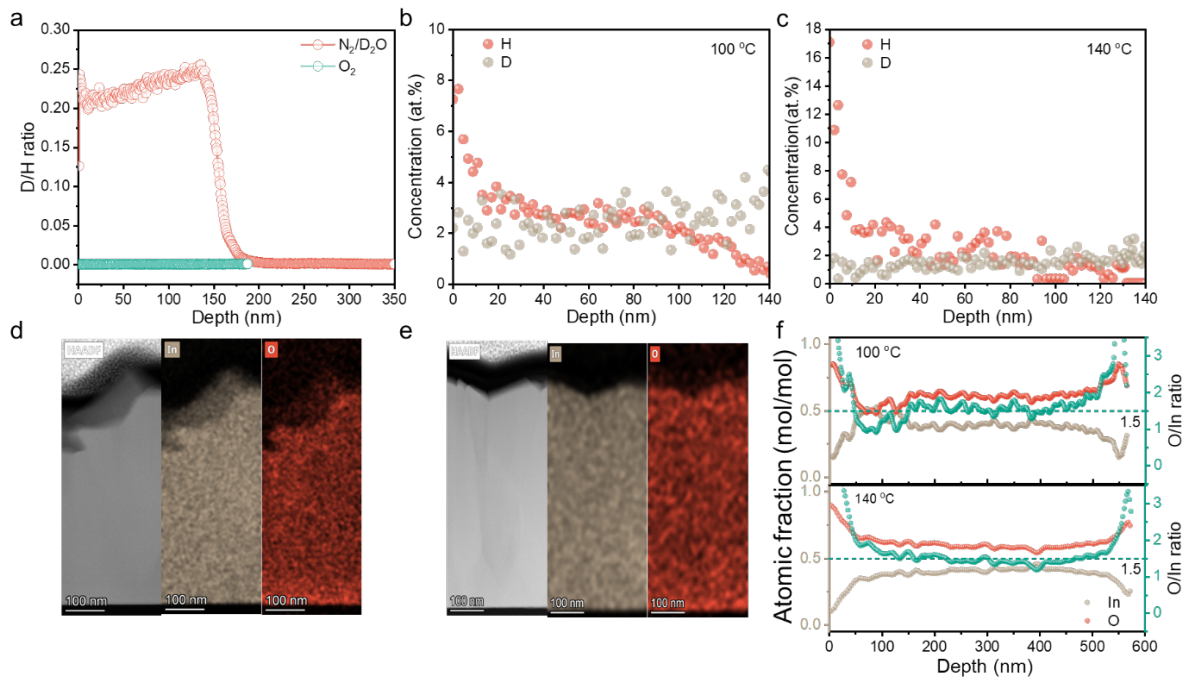


Figure 3. Understanding the mechanism of H doping in AP-CVD $\text{H}:\text{In}_2\text{O}_3$. **a.** Depth profiles of the D/H ratio in $\text{H}:\text{In}_2\text{O}_3$ thin films (thickness ~ 150 nm) deposited onto Si wafer at 100 °C using $\text{N}_2/\text{D}_2\text{O}$ or O_2 -

only in the oxidant channel, obtained from Nano-SIMS measurements. Depth profiles of D and H atomic concentrations in H:In₂O₃ thin films (thickness ~150 nm) deposited onto Si wafer at **b.** 100 °C and **c.** 140 °C using N₂/D₂O in the oxidant channel, obtained from ToF-ERDA measurements. STEM-EDX cross-section images and elemental (In, O) mappings of H:In₂O₃ films (thickness ~500 nm) deposited at **d.** 100 °C and **e.** 140 °C. **f.** Depth profiles of the atomic fractions of In and O, along with the O/In ratio, for H:In₂O₃ thin films deposited at 100 °C and 140 °C.

To quantify the distribution of hydrogen, we performed time-of-flight elastic recoil detection analysis (ToF-ERDA) measurements on N₂/D₂O-grown thin films at both 100 °C and 140 °C, as indicated in Figure 3b-c. While Nano-SIMS provides relative isotopic ratios, ToF-ERDA enables absolute quantification of light elements, allowing direct assessment of hydrogen incorporation and its dependence on deposition temperature [68]. The deuterium concentration remains relatively uniform throughout the film thickness, confirming homogeneous incorporation during growth, supporting a CVD-like hydrogen incorporation mechanism in which water-derived hydrogen is incorporated during each deposition cycle. In contrast, the hydrogen signal decreases near the film surface and exhibits an apparent reduction beyond ~90–100 nm depth. This reduction is attributed to decreased recoil detection efficiency for hydrogen at greater depths, rather than a true absence of hydrogen, as light elements such as hydrogen possess limited recoil energy in ToF-ERDA measurements. After renormalization excluding substrate-derived silicon contributions, the total hydrogen content (H + D) is comparable for both deposition temperatures (~4.8 at.%). This indicates that the overall hydrogen incorporation level is not strongly temperature dependent within the investigated range (**Table S6**, SI). However, the relative contribution of oxidant-derived hydrogen (deuterium) decreases at 140 °C compared to 100 °C. At 100 °C, the D concentration is higher, indicating that hydrogen incorporated from the water oxidant dominates. At 140 °C, although the total hydrogen content in the film remains similar as at 100 °C, the reduced D fraction and increased H fraction suggest a shift in hydrogen incorporation pathway, with a greater relative contribution from metal precursor-derived or background hydrogen sources. This trend is consistent with enhanced precursor decomposition at elevated temperature [69]. Carbon concentration also increases at 140 °C (Figure S9, SI), supporting increased thermal decomposition of the InCp precursor. Carbon-containing fragments may modify the local bonding environment and influence hydrogen chemical state, potentially affecting dopant activation efficiency. Importantly, when considered together with Table 1 and the XPS analysis above, these results suggest that hydrogen originating from the water oxidant is most closely associated with the improved electrical properties. Films grown with water-containing oxidant environments (O₂/H₂O or N₂/H₂O) exhibit higher carrier concentrations and lower oxygen-vacancy content, consistent with water-derived hydrogen acting both as a shallow donor and as a passivating species that suppresses oxygen-vacancy-related scattering. This provides a consistent explanation for the simultaneous increase in carrier concentration and conductivity observed for the H₂O-grown films.

To further assess the influence of oxidizing environment on overall composition, ToF-ERDA measurements were also performed on films deposited under different oxidants at 140 °C (Table S7, SI). Films grown under O₂-only and O₂/N₂ conditions exhibit significantly higher carbon contamination (3.3% and 1.9%, respectively) compared to N₂/H₂O and O₂/H₂O films (0.2% and 0.6%). Although elevated temperature promotes precursor fragmentation, the oxidant chemistry determines whether Cp-derived fragments are efficiently removed or retained within the growing film. Under O₂-only conditions, rapid surface oxidation may lead to non-equilibrium growth and incomplete ligand removal, resulting in increased carbon incorporation. In contrast, H₂O-containing environments appear to facilitate more effective ligand elimination, yielding lower residual carbon content. Carbon impurities are known to introduce mid-gap states that increase sub-bandgap absorption and degrade optical transmittance [69]. The reduced carbon contamination observed for N₂/H₂O and O₂/H₂O films is therefore consistent with their improved transmittance and reduced optical losses (Figure 2a).

The In and O distribution was investigated via Scanning Transmission Electron Microscopy-Energy Dispersive X-ray analysis (STEM-EDX). TEM-EDX elemental mapping of the cross-section of the H:In₂O₃ films deposited at 100 and 140 °C shows that both In and O are uniformly distributed across the film thickness, confirming good compositional homogeneity (Figure 3d-e). For the film deposited at 140 °C, the O/In ratio remained close to the stoichiometric value of 1.5 throughout the lamella, with small deviations at the film surface and at the film–substrate interface, which can be attributed to oxygen adsorption from air and the presence of SiO_x at the Si substrate (Figure 3f). Overall, the combined Nano-SIMS, ToF-ERDA, and STEM-EDX analyses demonstrate that hydrogen in AP-CVD-grown H:In₂O₃ originates predominantly from the water oxidant, is incorporated uniformly during growth, and is strongly influenced by oxidant chemistry and deposition temperature. These factors collectively govern defect passivation, stoichiometry, and ultimately the electrical and optical properties of the films.

Conclusions

In conclusion, we have demonstrated the rapid, atmospheric growth of high figure-of-merit TCOs using AP-CVD. We achieved phase-pure thin films of H:In₂O₃ (cubic bixbyite phase, with space group *Ia* $\bar{3}$) at a deposition temperature of only 140 °C, without requiring any ozone or plasma. By investigating in detail the effects of the oxidants used, we found that using H₂O vapor with N₂ dilution gas resulted in higher-performance H:In₂O₃ than using O₂ gas as the oxidant. Isotope labelling experiments (using D₂O), together with nano-SIMS and ToF-ERDA measurements reveal the benefits of H₂O as the oxidant by introducing H as shallow donor dopants, which increases the carrier concentration, in contrast to the more passivate role of H introduced from the InCp precursor. These H dopants may also passivate oxygen vacancies. This is consistent with XPS measurements that show a lower oxygen vacancy concentration in the films grown with H₂O rather than O₂. The mobility of these films improved from

$40\pm 10\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ (using O_2) to $160\pm 30\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ (using H_2O), which is consistent with reduced carrier scattering from oxygen vacancies. Furthermore, by increasing film thicknesses from $470\pm 40\text{ nm}$ to $710\pm 50\text{ nm}$, we increased mobilities further to $190\pm 30\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$. Given the high mobilities, we achieved low sheet resistance down to $7.20\pm 0.01\text{ }\Omega\text{ }\square^{-1}$ ($0.50\pm 0.06\text{ m}\Omega\text{ cm}$ resistivity) without requiring excessively high carrier concentrations, thus avoiding high free carrier absorption. This allowed us to maintain high NIR transmittance (89%) as well as high visible transmittance (85.5%), outperforming commercial ITO ($13\pm 1\text{ }\Omega\text{ }\square^{-1}$ sheet resistance, 67.8% NIR transmittance). Furthermore, despite the mild growth conditions, we achieved $40\times$ higher growth rate than ALD ($0.19\pm 0.01\text{ nm s}^{-1}$ vs. $\sim 0.005\text{ nm s}^{-1}$), such that the total processing time required to achieve $0.2\text{--}0.8\text{ m}\Omega\text{ cm}$ resistivity $\text{H:In}_2\text{O}_3$ films was lowered from $>160\text{ min}$ (ALD) to only $40\pm 2\text{ min}$ with AP-CVD. This is competitive with sputter deposition of ITO ($25\pm 5\text{ min}$), but with AP-CVD, we have the advantages of fully atmospheric processing (which could enable lower capital intensity of manufacturing) as well as gentler deposition, which is essential for growth onto soft semiconductors. Overall, this work establishes AP-CVD as an appealing method for the scalable manufacture of TCOs that addresses the processability-performance gap of existing deposition methods.

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Author contributions

X. G., and R. L. Z. H. conceived of the project. X. G. deposited all H:In₂O₃ thin films using atmospheric pressure chemical vapor deposition and analyzed all data collected. X. G. measured optical and electrical properties with the assistance of J. O., who is under the supervision of R. S. B. H-J. S. and H-K. K. helped with ellipsometer measurement. E. L. Q., M. K. S. and C. D. M. conducted Time-of-flight elastic recoil detection analysis. Y-T. H. helped with the scanning electron microscope measurement. X. L. conducted scanning transmission electron microscopy- energy dispersive X-ray analysis under the supervision of C. D. K. L. conducted Nano-SIMS analysis under the supervision of K. L. M. C.-Y. C. and Y. W. contributed to discussions of potential applications of the thin films. A. S. contributed to conceive the project. R. L. Z. H. supervised the overall work. All authors contributed to the writing and editing of the manuscript.

Conflicts of interest

R. L. Z. H. declares a competing financial interest, in that he is Chief Technology Officer of *NanoPrint Innovations Ltd.*, a company manufacturing spatial atomic layer deposition equipment.

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